

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the properties of water	To include the polar nature of the water molecule, hydrogen bonding and the role of water as a solvent.
(b) (i) the importance of water as a major constituent of cells and body fluids, and as the essential transport medium in plants and animals	To include the transpiration stream in plants, and plasma, serum, tissue fluid, lymph and urine in mammals. <i>M0.1, M0.2, M0.3, M1.1, M1.2, M1.6, M3.1</i> HSW5, HSW6
(ii) analysis of secondary data on the composition of mammalian body fluids to illustrate the role of water as a solvent	To include solutes (sugars, proteins), electrolytes (hydrogen ions, H^+ , potassium ions, K^+ , sodium ions, Cl^- , hydrogencarbonate ions, HCO_3^-).
(c) (i) how sugar and protein molecules in body fluids can be detected and measured	To include the use of reagent test strips and biosensors to detect and measure the concentration of sugars and proteins. <i>M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M3.1, M3.2</i> PAG5, PAG9 HSW3, HSW4, HSW5, HSW6
(ii) the methodology and interpretation of the results of the Biuret test, Benedict's test and colorimetry	
(d) the importance of hydrolysis and condensation of biological molecules in cell metabolism	To include the concept of monomers and polymers in a range of biological molecules.
(e) the structure of the ring form of α -glucose as an example of a simple monosaccharide, and lactose as a disaccharide	
(f) (i) the formation of polysaccharides by condensation	To include glycogen and starch (amylose and amylopectin) AND the formation of 1,4- and 1,6-glycosidic bonds and reference to the significance of branching on solubility.
(ii) a test for the identification and measurement of starch	To include the qualitative test for starch using iodine and colorimetry. PAG5, PAG9

(g) osmosis, in terms of the effect of solutes and electrolytes on the water potential of body fluids

To include the effect of concentration of plasma proteins, glucose and electrolytes on the water potential of blood.

M0.3, M1.1, M1.2, M1.3, M3.1, M3.2, M3.4

(h) practical investigation(s) into factors affecting osmosis in plant and animal cells.

M0.3, M1.1, M1.2, M1.3, M3.1, M3.2, M3.4

PAG8

HSW3, HSW4, HSW5, HSW6

Draft